

SDS
 → Fundamentals
 → Probability
 → Information Theory
 APS & ML

Inf. Ig.
 Entropy
 Joint entropy
 Relative Ent.
 KL div.

loss funcⁿ ← **CE**
 Solving linear systems
 $Ax = b$
 $\Rightarrow x = ?$

MfDS
 Cov. Mat.
 ↳ symmetric
 ↳ PSD

Linear Algebra:
 Eigen values ↳ LI
 Eigen vectors ↳ PSD
 ↳ PD matrix
 Matrix Decomposition

Optimization
 Convex & Concave Set
 Convex & Concave function
 Minima, Maxima
 Gradient

SGD
 ↳ #

Langrangian method

SVM

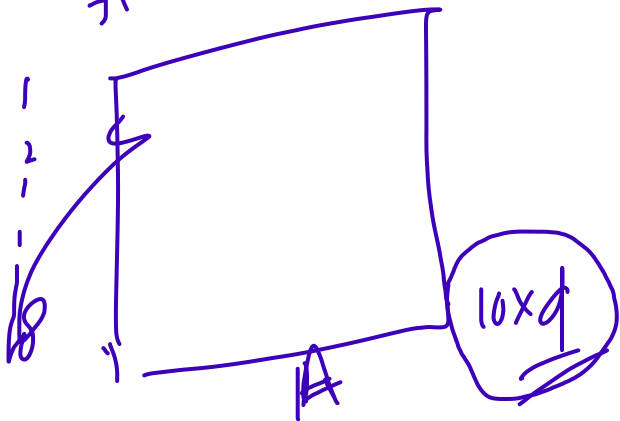
Vectors
 ↳ Matrix
 ↳ Norm
 ↳ Orthonormal LI
 ↳ orthonormal vectors
 NMI
 DNN
 PCA
 Applications
 FDR

Eigen Decomposition
 SVD
 ↳ full SVD
 ↳ Reduced SVD
 ↳ Applications

* Linear regression

Multivariate

$f_1, f_2, \dots, f_d, y$



$$x_{d \times 1} = y_{10 \times 1}$$

$$x = \begin{pmatrix} -1 \\ A \end{pmatrix} y$$

$$\underline{\underline{d=10}}$$

- ① LSA
- ② MN SMⁿ

Scalar
Vector
Matrix
Tensor

image / video / text
or
matrix
Vector





Linear Algebra

CNN

video, color image $\in \mathbb{R}, h, b$
Input & Output $\rightarrow$ vectors
matrices

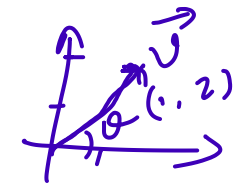
scalar, vector, Matrix, Tensor
 $d \in \mathbb{R}$ $x \in \mathbb{R}^d$ $D \in \mathbb{R}^{N \times d}$ $\mathbb{R}^{d \times d \times m}$
 $W \in \mathbb{R}^{n \times n}$

Vector in $\mathbb{R}^n$

$$V = [v_1, v_2, \dots, v_n]^T \in \mathbb{R}^n, v_i \in \mathbb{R}$$

$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$n=2, v=[1,2]$



$v_1 = (1,3), v_2 = (1,1)$

$v_1 + v_2 = (2,4)$

Addition/subtraction: ✓

Dot Product:

$v_1 = (x_1, x_2, \dots, x_n)$

$v_2 = (y_1, y_2, \dots, y_n)$

$v_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} v_2 = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$

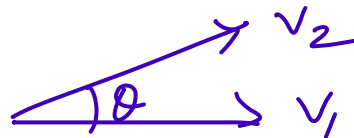
$$v_1 \cdot v_2 = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

$$= \sum_{i=1}^n x_i \cdot y_i = \text{Simple vec}$$

$d = v_1^T \cdot v_2 = v_2^T \cdot v_1$
 $(1 \times 3) \times (3 \times 1) = 1 \times 1$

Magnitude: $|v| = \sqrt{x^2 + y^2 + \dots + z^2}$

Angle b/w 2 vectors:



$$v_1 \cdot v_2 = |v_1| |v_2| \cos \theta$$

$$\cos \theta = \frac{v_1 \cdot v_2}{|v_1| |v_2|} \Rightarrow \theta = ?$$

Linear Combination of Vectors:

* set $S = \{v_1, v_2, v_3, \dots, v_k\}$, then a new vector

$$v = d_1 v_1 + d_2 v_2 + \dots + d_n v_n = \sum_{i=1}^n d_i v_i$$

linear combination of set of vectors is LI & LI?

LI or LI vectors:

$S = \{v_1, v_2, \dots, v_k\}$ is LI if

$$d_1 v_1 + d_2 v_2 + \dots + d_n v_n = 0 \text{ holds only when } d_1 = d_2 = \dots = d_n = 0$$

① $S = \{ \underset{v_1}{(1,0)}, \underset{v_2}{(1,1)} \}$

$$d_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + d_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0 \text{ when } d_1 = d_2 = 0$$

$$\Rightarrow d_1 + d_2 = 0 \quad \checkmark \quad \text{LI}$$

$$d_2 = 0 \quad \checkmark$$

$\{ (1,0), (0,1) \} \Leftrightarrow \mathbb{R}^2$

Basis

$\hookrightarrow$ LI



Basis vectors:

Span:

Set of all vectors that can be obtained

$\mathbb{R}^3$
 $\{ (1,0,0), (0,1,0), (0,0,1) \}$



Orthogonal vector:

$\perp$, dot product is zero

A set of vectors $\{v_1, v_2, \dots, v_n\}$ are mutually orthogonal if $v_i \cdot v_j = 0 \quad \forall i \neq j$

Ex: $\{ \underbrace{(1, 0, -1)}_{v_1}, \underbrace{(1, \sqrt{2}, 1)}_{v_2}, \underbrace{(1, -\sqrt{2}, 1)}_{v_3} \} \in \mathbb{R}^3$

Orthogonal vectors:

→ mutually orthogonal

→ magnitude $\rightarrow 1$

Ex: $\{ \underbrace{(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})}_{v_1}, \underbrace{(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})}_{v_2} \}$

non-zero
orthogonal $\rightarrow 2i$
vector

L^1

$\rightarrow$ orthogonal (Gram-Schmidt Algorithm)

$$\frac{v_1}{|v_1|} = \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}} \right)$$

$$\frac{v_2}{|v_2|} = L$$

$$\frac{v_3}{|v_3|} = L$$

Matrix:

$$\begin{bmatrix} a_{ij} \end{bmatrix}_{m \times n} \left(\begin{array}{c} \\ \\ \end{array} \right)_{m \times n}$$

Upper triangular matrix

$$\begin{bmatrix} 5 & 8 & 7 \\ 0 & 3 & 8 \\ 0 & 0 & 1 \end{bmatrix}$$

Diagonal Matrix: $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{\text{zero } m \times n}$$

Identity Matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2}$

Addition: A & B size must be same
A + B
A - B

Subtraction:

element-wise

Commutative $A + B = B + A$

Associative $A - B \neq B - A$

$$A + (B + C) = (A + B) + C$$

Scalar multiplication

2A

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

$$2A = \begin{bmatrix} & \\ & \end{bmatrix}_{2 \times 2}$$

broader:

concept of matrices

$$A + V = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix}$$

2×3

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3} + \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}_{1 \times 3} = V$$

Element-wise multiplication

$$A \cdot B$$

✓
Size should be same

Multiplication

$$A \times B = C$$

$p \times q$ $q \times r$ $p \times r$

$$b \times p$$

$p \times p$ $p \times p$

$$AI = I \cdot A = A$$

$$A \times B \neq B \times A$$

$$A \times B = 0 \text{ if } A=0 \text{ or } B=0$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

A B $A \times B = 0$ $B \neq 0$

$$\begin{cases} x \neq 0 \\ y = 0 \\ x = 0 \\ y = 0 \end{cases}$$

Transpose:

$$A = \{a_{ij}\}_{n \times n} \quad \begin{matrix} i = 1, \dots, m \\ j = 1, \dots, n \end{matrix}$$

$$A^T = \{a_{ji}\}_{n \times m}$$

Prop:

$$(A+B)^T = A^T + B^T$$

$$(AB)^T = B^T A^T$$

$$(kA)^T = k A^T$$

$$(A^T)^T = A$$

Pseudo-inverse
 $[A_{m \times n}]^{-1} \rightarrow [A_{n \times n}]^{-1}$
reverses

Symmetric Matrix:

$$A^T = A$$

Determinant of a matrix

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}_{2 \times 2}$$

$$|A| \text{ or } \det(A) = 3 - (-2) = 5$$

Inverse of a matrix:

$$\text{scalar } k = 7, \quad k^{-1} = \frac{1}{7}$$

if the determinant is non-zero
→ Singular matrix
→ Non-singular "

prop:

$$AA^{-1} = A^{-1}A = I$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$(A^{-1})^{-1} = A$$

$$(A^T)^{-1} = (A^{-1})^T$$

$$(kA)^{-1} = \frac{1}{k} A^{-1}$$

Trace:

$$A = \begin{bmatrix} 7 & 4 & 3 \\ 7 & 2 & 6 \\ 8 & 9 & 3 \end{bmatrix}_{3 \times 3}$$

$$\begin{bmatrix} \cdot & \cdot & \cdot \\ & \cdot & \cdot \\ & & \cdot \end{bmatrix}$$

$$\text{Tr}(A) = 6$$

Orthogonal Matrix:

$$A^T = A^{-1}$$

$$A^T A = A A^T = I$$

Norm:

Tensor $\xrightarrow{\text{Norm}(\cdot)}$ Scalars
(vector/matrix)

Norm is any function f that satisfies

① $f(x) = 0 \quad x = 0$

② $f(x+y) \leq f(x) + f(y)$ (Triangle inequality)

③ $\forall \alpha \in \mathbb{R}, f(\alpha x) = |\alpha| f(x)$ (Linearity)

$\alpha = 2, v = (1, 2, -3)$
 $\|2v\| = ?$

$w = (2, 4, -6)$
 $\sqrt{2^2 + 4^2 + (-6)^2} = \sqrt{56} = ?$

$2\|v\| = 2 \times \sqrt{1^2 + 2^2 + (-3)^2} = 2\sqrt{14} = ?$

L^2 -norm

Types: $v = (v_1, v_2, \dots, v_n)$
 ① Euclidean norm. $\|v\|_2 = (v_1^2 + v_2^2 + \dots + v_n^2)^{\frac{1}{2}}$ $\swarrow$ 2-norm
 $\nwarrow$ $p=2$

② 1-norm: $\|v\|_1 = |v_1| + |v_2| + \dots + |v_n|$ $\nwarrow$ $p=1$
 $\nearrow$ $p > 1$

p -norm $\Rightarrow \|v\|_p = (|v_1|^p + |v_2|^p + \dots + |v_n|^p)^{\frac{1}{p}}$

③ ∞ -norm or max-norm: $\|v\|_\infty = \max(|v_1|, |v_2|, \dots, |v_n|)$
 $v = (-5 \ 3 \ 2)$

$$\|v\|_2 = \sqrt{(-5)^2 + 3^2 + 2^2} = \sqrt{38}$$

$$\|v\|_1 = 5 + 3 + 2 = 10$$

$$\|v\|_\infty = 5$$

Frobenius norm $A_F = \left(\sum_{ij} A_{ij}^2 \right)^{\frac{1}{2}}$

$$= \sqrt{\begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}^2} = \sqrt{1^2 + 2^2 + 2^2 + 0^2}$$

Eigen values & Eigen vectors:

③ $\text{Trace}(A) = \text{sum of eigen values}$
④ $\text{Det}(A) = \text{product of eigen values}$

Properties:

$$A_x = x + 1$$

① $A^{-1}x = \frac{1}{\lambda}x$

$$A \times \frac{1}{2} = 7^2$$

② $A^2 x = \lambda^2 x$

$$A^k x = x^k \quad x, k \in \mathbb{Z}$$

$$Av = \lambda v$$

$$|A - \lambda I| = 0$$

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\lambda = 2, 1$$

$$v = 1.$$

⑤ Eigen vectors with distinct eigenvalues are L.I.
 $\Sigma = \{ \vec{v}_1, \vec{v}_2, \dots, \vec{v}_n \}$

are LI.

~~$\Sigma = \|Ax - b\|_2^2$~~

(b) for symmetric α 's are

6) for symmetric matrix, $x = \begin{bmatrix} 1 \\ 3 \end{bmatrix}_2$
 matrix, e_i are
 orthogonal to each other
 eigenvalues are
 real & +ve. A
 $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$x = \begin{bmatrix} 1 \\ 3 \end{bmatrix}_{2 \times 1} \quad A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}_{2 \times 2}$$

$$A^* = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$y = Ax = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A^T =$$

Angewandte Multiplikation

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

$$\lambda = 2, 1, 1$$

AM $2^2 = 1$
 $1 = 2$

3×3
 $\lambda_1 =$
 $\lambda_2 =$
 $\lambda_3 =$

 dirⁿ same